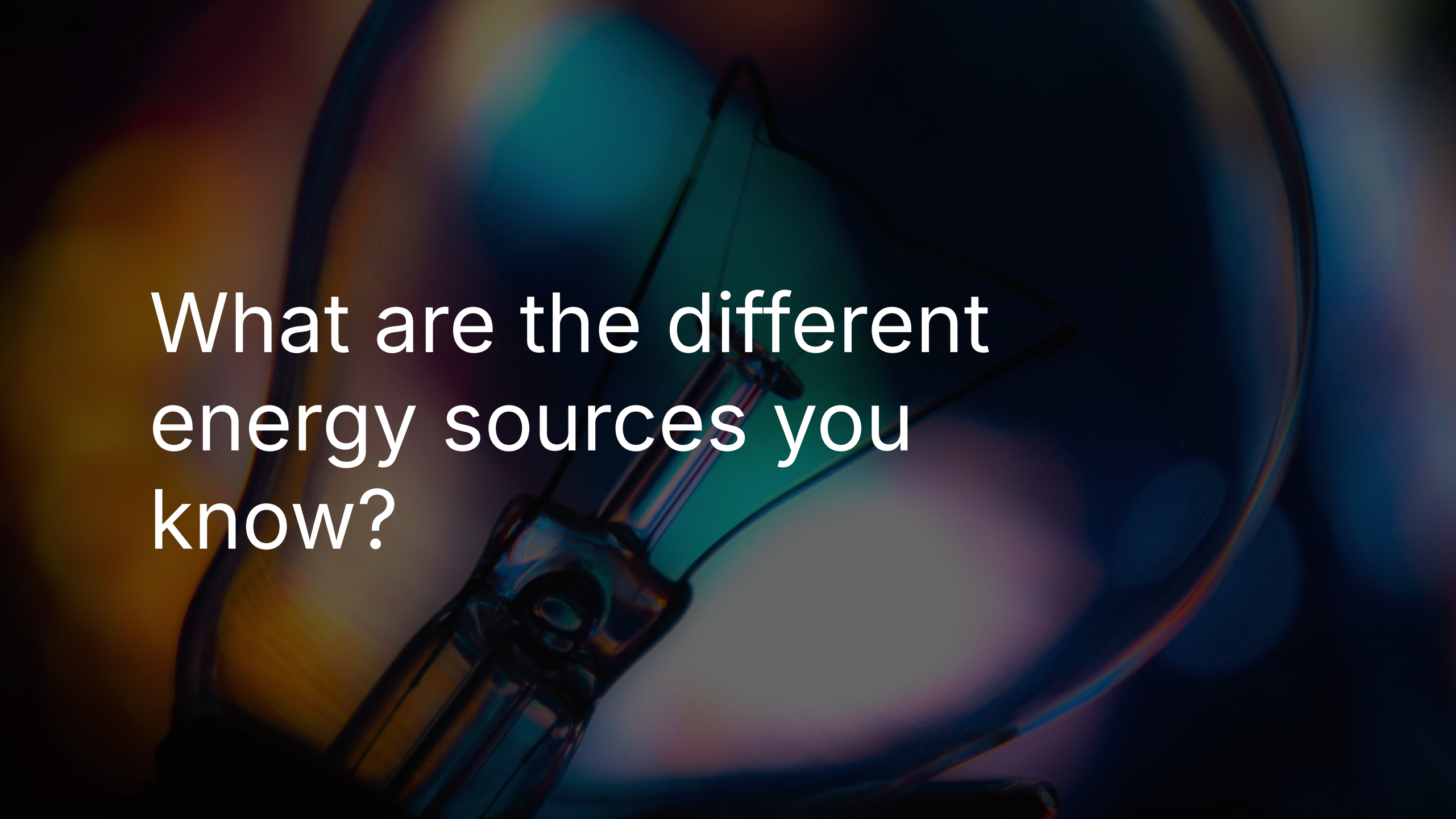
Energy:Past, Present, and Future

LESSON 10

Objectives

By the end of the lesson, the students should be able to:

1. Describe the major sources and uses of energy in the past, present, and future.
2. Identify the advantages and disadvantages of different energy sources.
3. Identify the environmental impact of energy use.

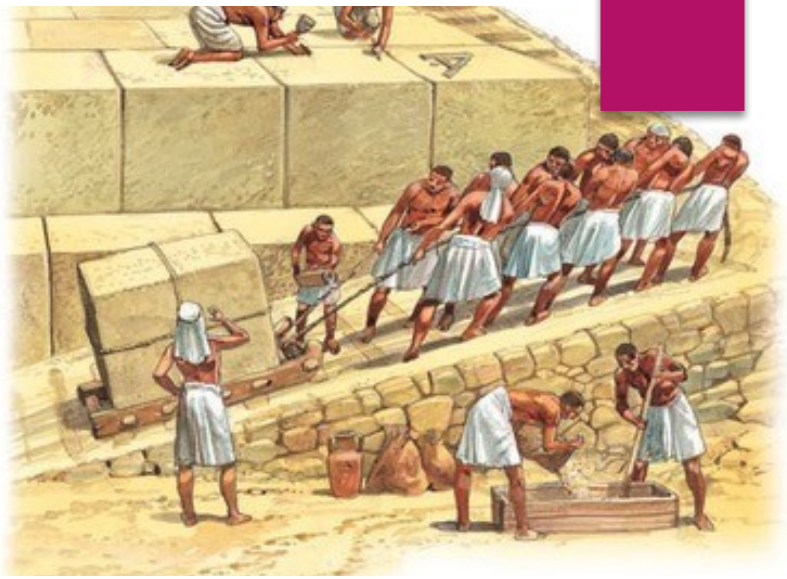

What are the different energy sources you know?



Energy-Past

Human and Animal Power

- **Human Energy**
  - Labor-intensive tasks such as farming, building, and transportation relied on human muscle power.
- **Animal Energy**
  - Animals like oxen, horses, and donkeys were used for plowing fields, transporting goods, and even turning simple machinery like mills.





Biomass

- **Wood**
  - The primary fuel for cooking, heating, and sometimes small-scale industrial processes.
- **Crop Residues and Dung**
  - In areas with limited access to wood, agricultural residues or dried animal dung were used as fuel.









Wind Power

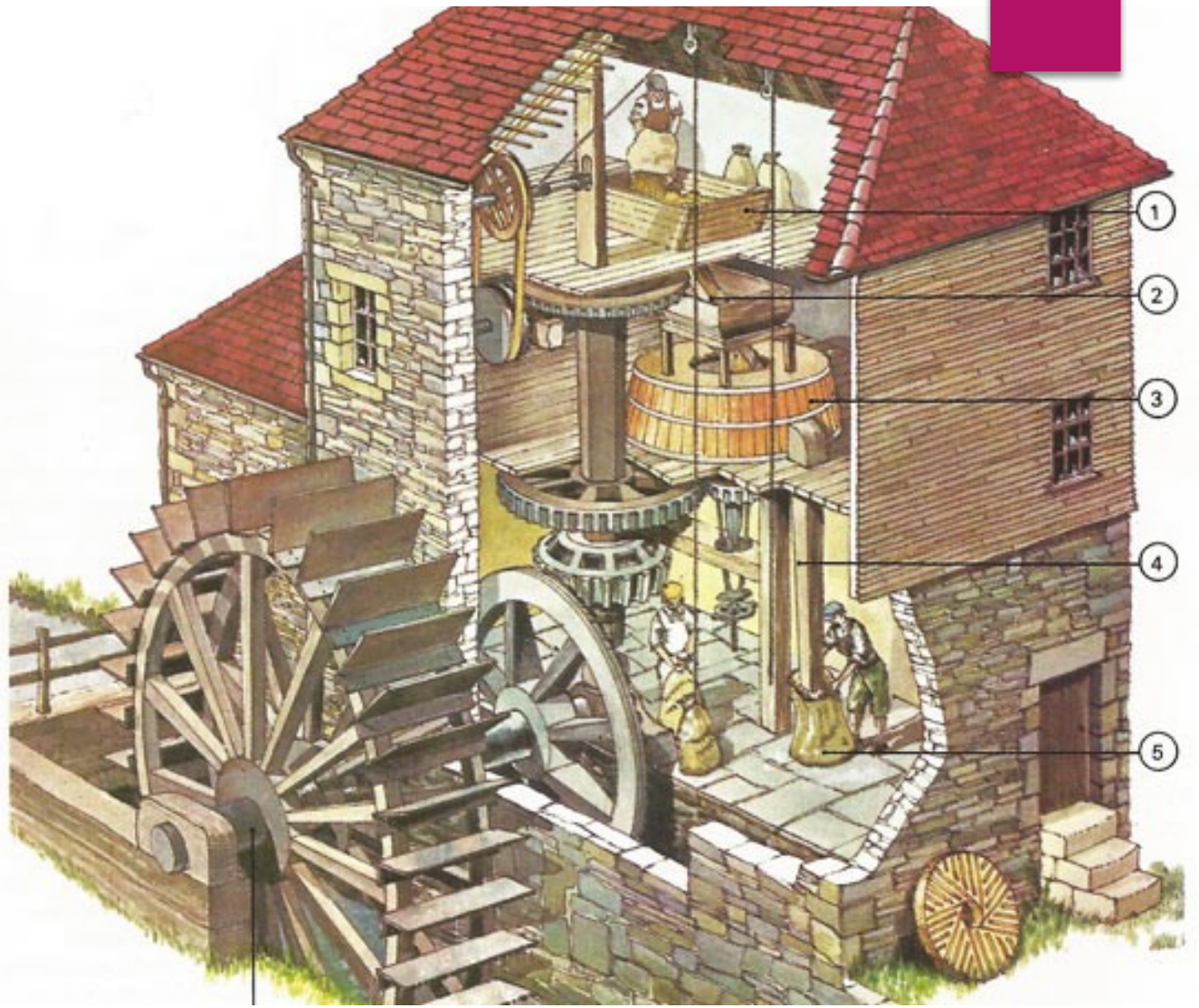
- Windmills
  - Used for grinding grains into flour or pumping water.
- Sailing
  - Wind-powered ships were essential for transportation and trade.





Water Power

- Water Wheels
  - Utilized for grinding grains, sawing wood, or powering simple industrial machines.

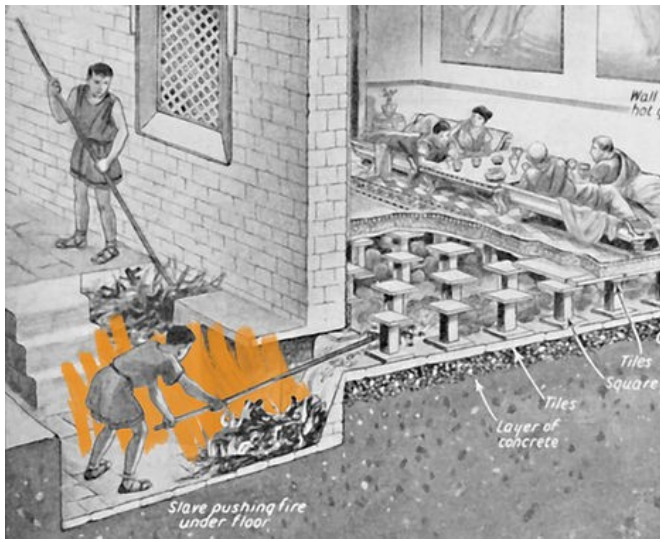


Early Use of Fossil Fuels

- **Coal**
  - In some regions, coal was used as early as the Roman period for **heating** and later for **blacksmithing**.
  - It became more prominent during the Industrial Revolution.







Sunlight

- Direct solar energy was used for drying crops, preserving food, and warming dwellings.







Energy: **Present** and **Future**



Energy sources

Non-renewable energy sources

- Fossil fuels (coal, oil, natural gas)
- Nuclear energy

Renewable energy sources

- Solar energy
- Wind energy
- Hydropower
- Geothermal energy
- Biomass energy

Fossil Fuels

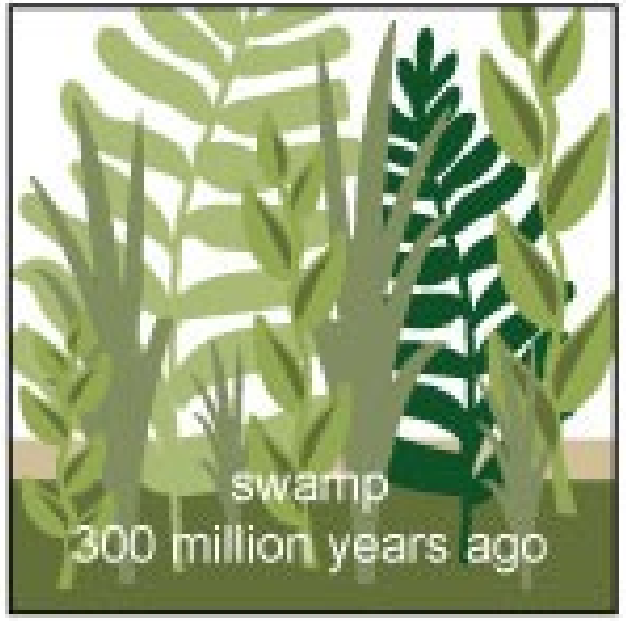
Fossil fuels are energy sources formed from the **remains of plants and animals that lived millions of years ago.** Over time, **heat and pressure** transformed these remains into fuels such as coal, oil (petroleum), and natural gas.

- **Coa**l: Formed from **plant matter** compressed over millions of years.
- **Oil (Petroleum)**: Formed from the **remains of marine organisms.**
- **Natural Gas:** Primarily composed of methane, formed from the **decomposition of organic matter.**

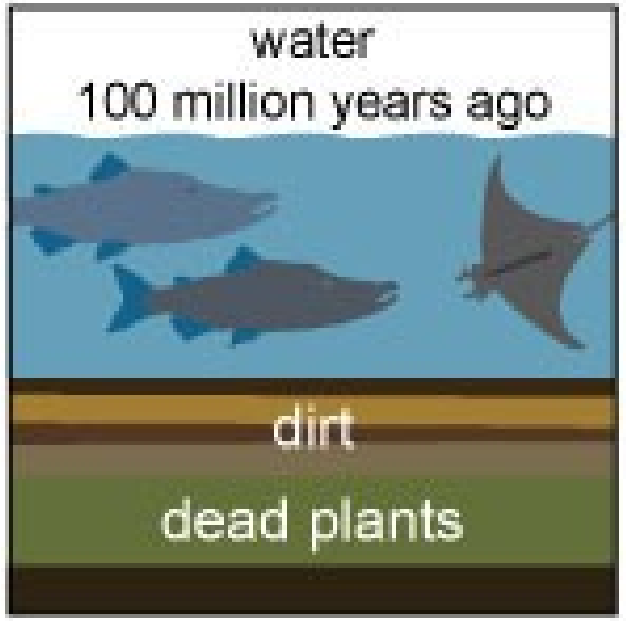


How coal was formed

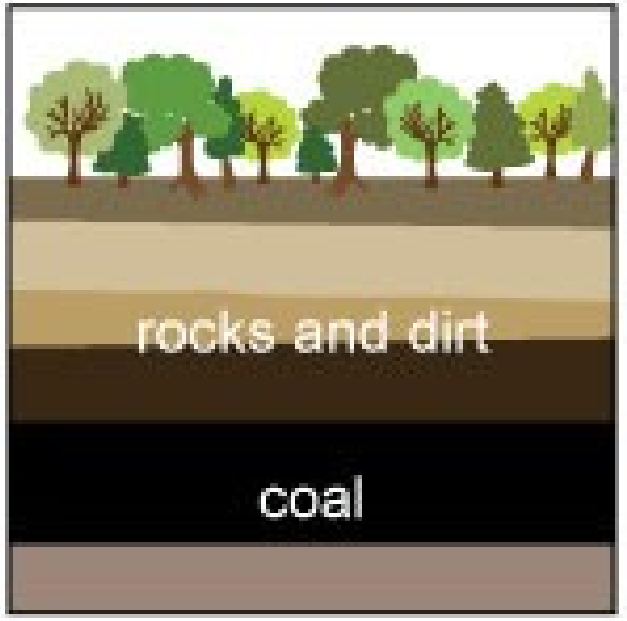
Before the dinosaurs, many giant plants died in swamps.



Over millions of years, the plants were buried under water and dirt.



Heat and pressure turned the dead plants into coal.



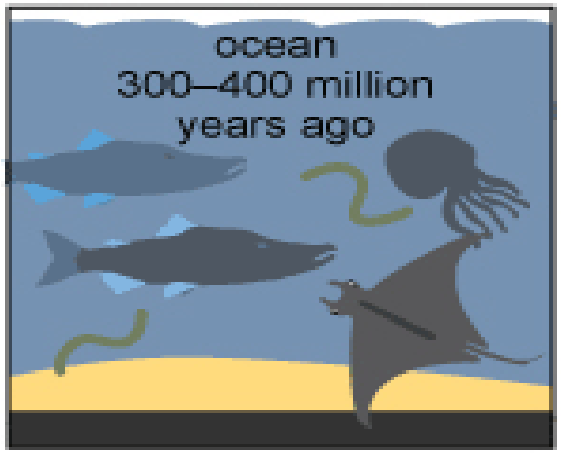
Source: Adapted from National Energy Education Development Project (public domain)

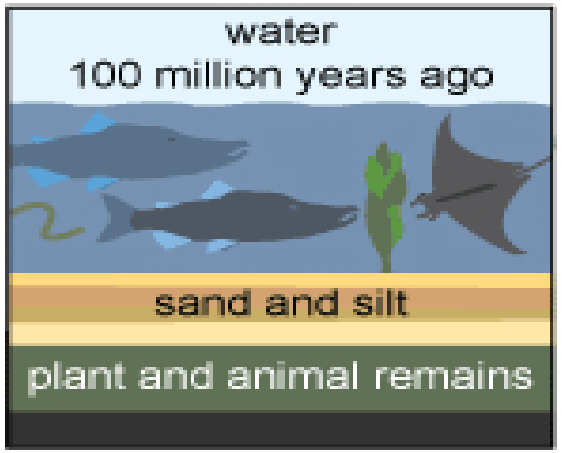
Petroleum and natural gas formation

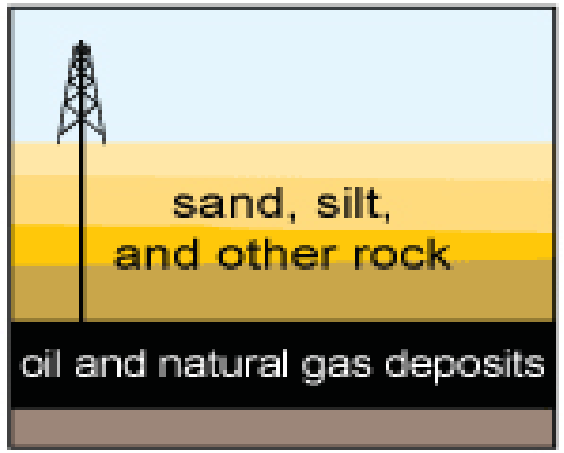
Tiny marine plants and animals died and were buried on the ocean floor. Over time, the marine plants and animals were covered by layers of silt and sand.

Over millions of years, the remains were buried deeper and deeper. The enormous heat and pressure turned the remains into oil and natural gas.

Today, we drill down through layers of sand, silt, and rock to reach the rock formations that contain oil and natural gas deposits.





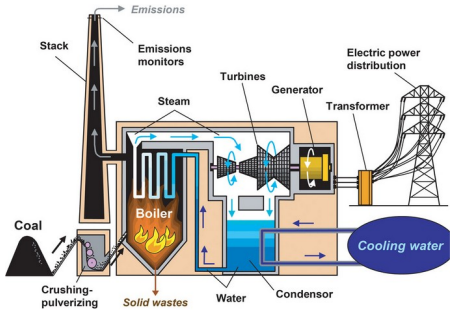


Data source: Adapted from National Energy Education Development Project (public domain)

Fossil Fuels

1. **Coal**
   - Used in power plants to generate electricity and in some industries for heating.
2. **Oil (Petroleum)**
   - Refined into gasoline or diesel to fuel cars, airplanes, and ships.
   - Used to make products like plastics and asphalt.
3. **Natural Gas**
   - Used in homes for cooking and heating.
   - Powers electric generators in some power plants.







Fossil Fuels

Advantages

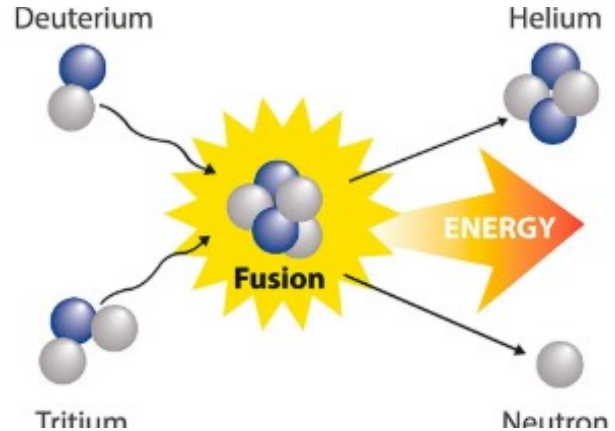
- Abundant and relatively inexpensive.
- Produces a large amount of energy from a small amount of fuel.
- Established infrastructure

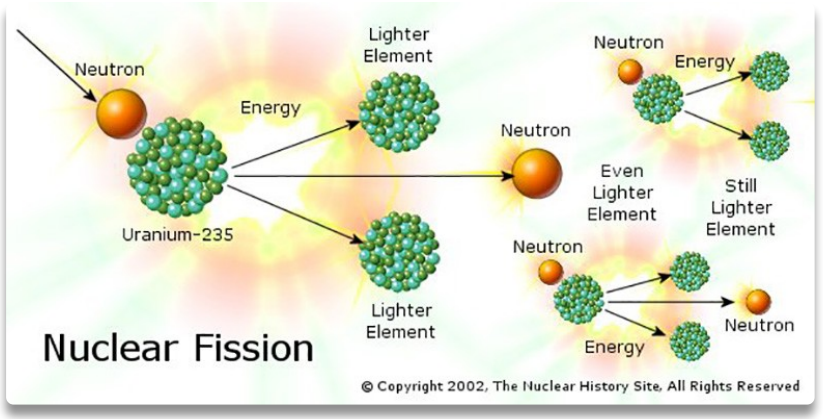
Disadvantages

- Contribute to air pollution and greenhouse gas emissions.
- **Finite** resource.
- Potential for accidents and spills.

Nuclear Energy

- **Nuclear Energy** is the **energy released during nuclear reactions**, either by combining **(fusion)** or through splitting **(fission)** of atomic nuclei.
- It is a powerful energy source commonly used to **generate electricity in nuclear power plants.**





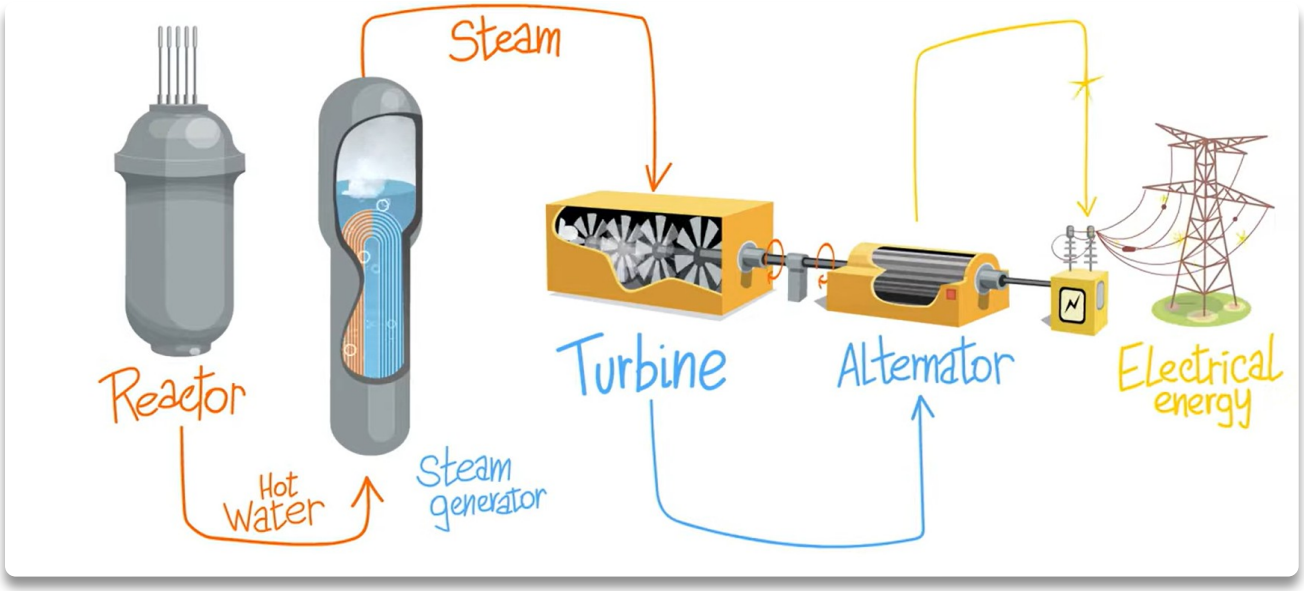
Nuclear Energy

Nuclear Fission

- The most common method in **power plants.**

Converting Heat to Electricity

- Nuclear fission
  - Atoms of uranium or plutonium are split, releasing a large amount of energy.
  - This energy heats water, which turns turbines to generate electricity.



How does a nuclear power plant work?



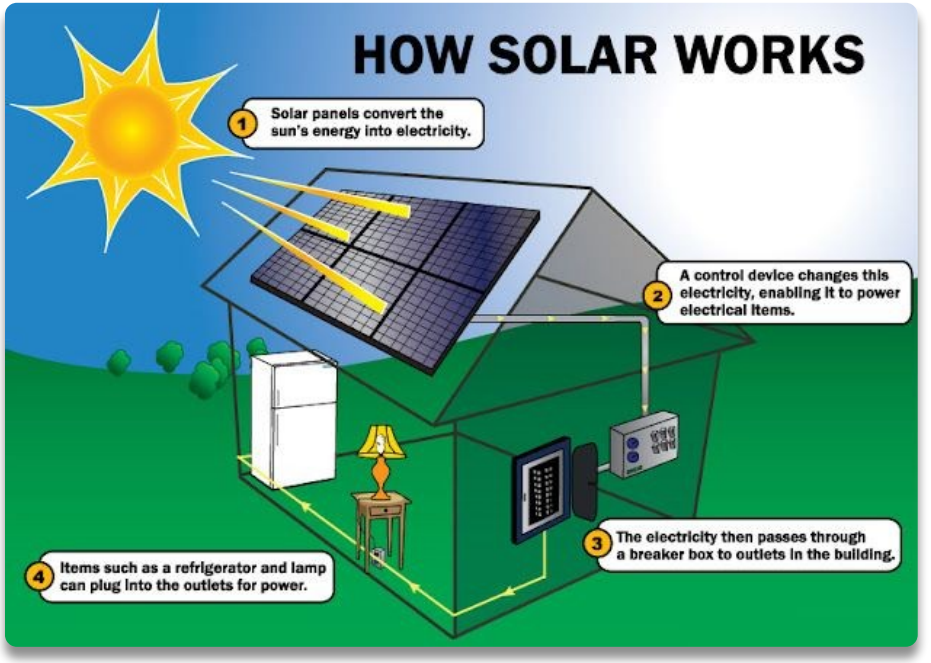
The Genkai nuclear power plant in southwestern Japan

Nuclear Energy

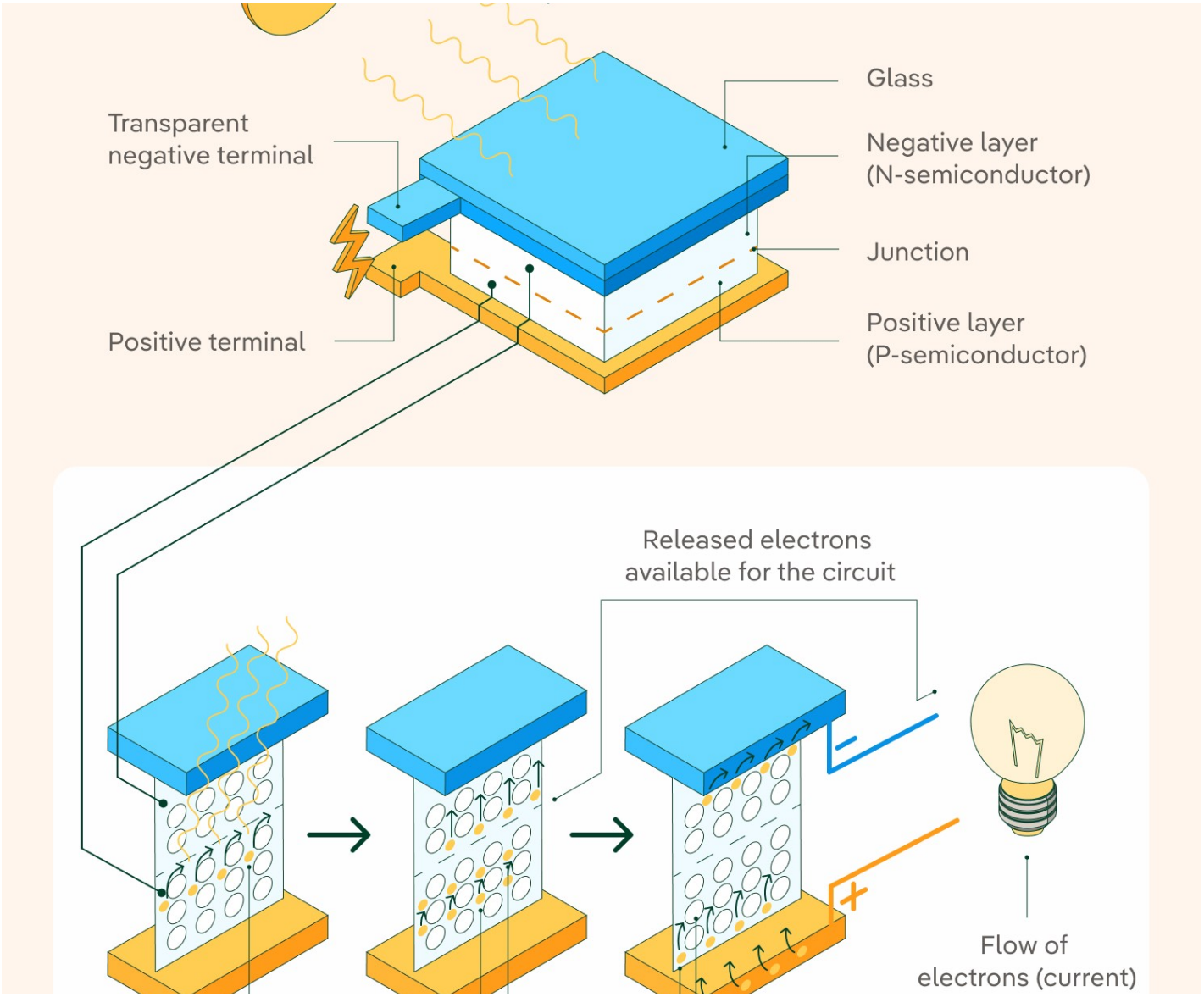
- **Advantages**
- Low greenhouse gas emissions.
- Produces a large amount of energy from a small amount of fuel.
- Reliable power source.
- **Disadvantages**
- Nuclear waste disposal
- Risk of accidents(like Chernobyl (1986) and Fukushima (2011)).
- High initial cost for building power plants

Solar Energy

- Solar panels convert **sunlight** into electricity through photovoltaic cells.
  - A photovoltaic cell (solar cell) is an electronic device that converts the energy in the solar radiation that reaches the earth in the form of light (photons) into electrical energy (electrons).
    - **photoelectric effect**: When sunlight hits the solar cell, the energy from the photons excites the electrons in the silicon, freeing them to flow as electric current, which is then harnessed as electrical energy.




photovoltaic cell (solar cell)

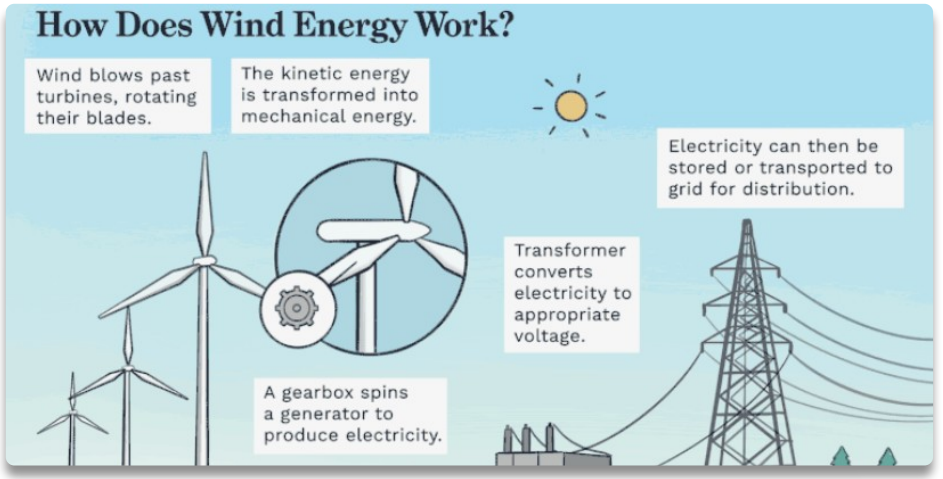

Transparent
negative terminal
Positive terminal
Glass
Negative layer
(N-semiconductor)
Junction
Positive layer
(P-semiconductor)
Released electrons
available for the circuit
Flow of
electrons (current)

Solar Energy

- **Advantages**
  - Clean and renewable.
  - Reduces reliance on fossil fuels.
  - Can be used for both electricity and heating.
- **Disadvantages**
  - Initial cost of installation
  - Weather-dependent
  - Requires large areas for efficient power generation

Wind Energy

- Wind turbines convert the kinetic energy of **wind** into electricity.
  - Kinetic energy is the energy an object has because of its motion.





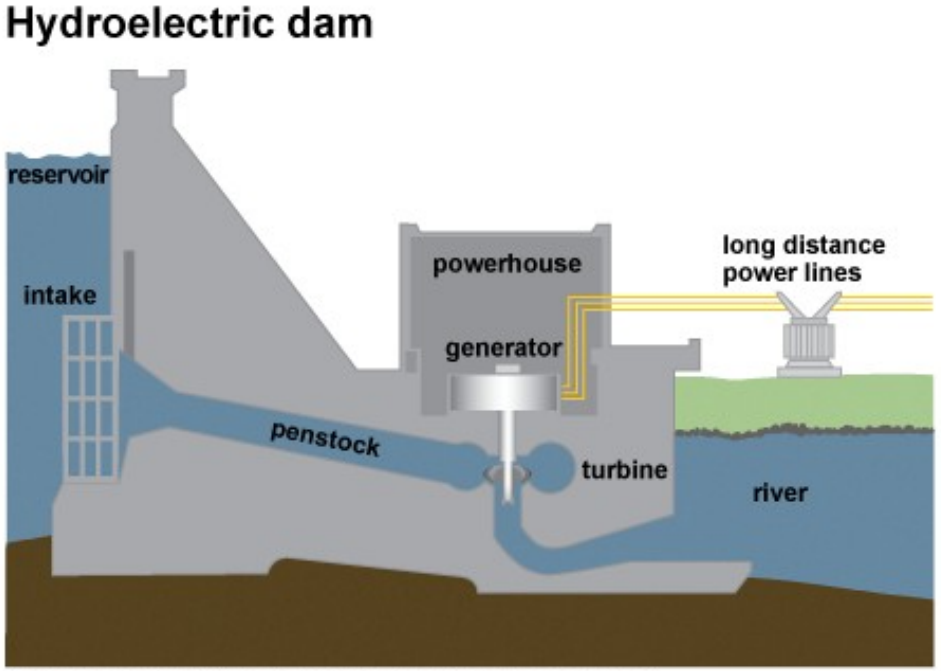
Wind Energy

- **Advantages**
  - Clean and renewable
  - Low operating costs
  - Minimal land use
- **Disadvantages**
  - Wind speed variability
  - Visual impact
  - Noise pollution

Hydropower

- Dams harness the potential energy of **water** to generate electricity.
  - Potential energy is the energy stored in an object due to its **position.**

- **Magat Hydro Plant** in Luzon has a capacity of 360 MW.
- **Angat Hydropower Plant** in Luzon has a capacity of 218 MW and supplies Manila with 90% of its fresh water.
- **Agus VI Hydroelectric Plant** in Mindanao has a capacity of 200 MW and has been online for over 50 years.



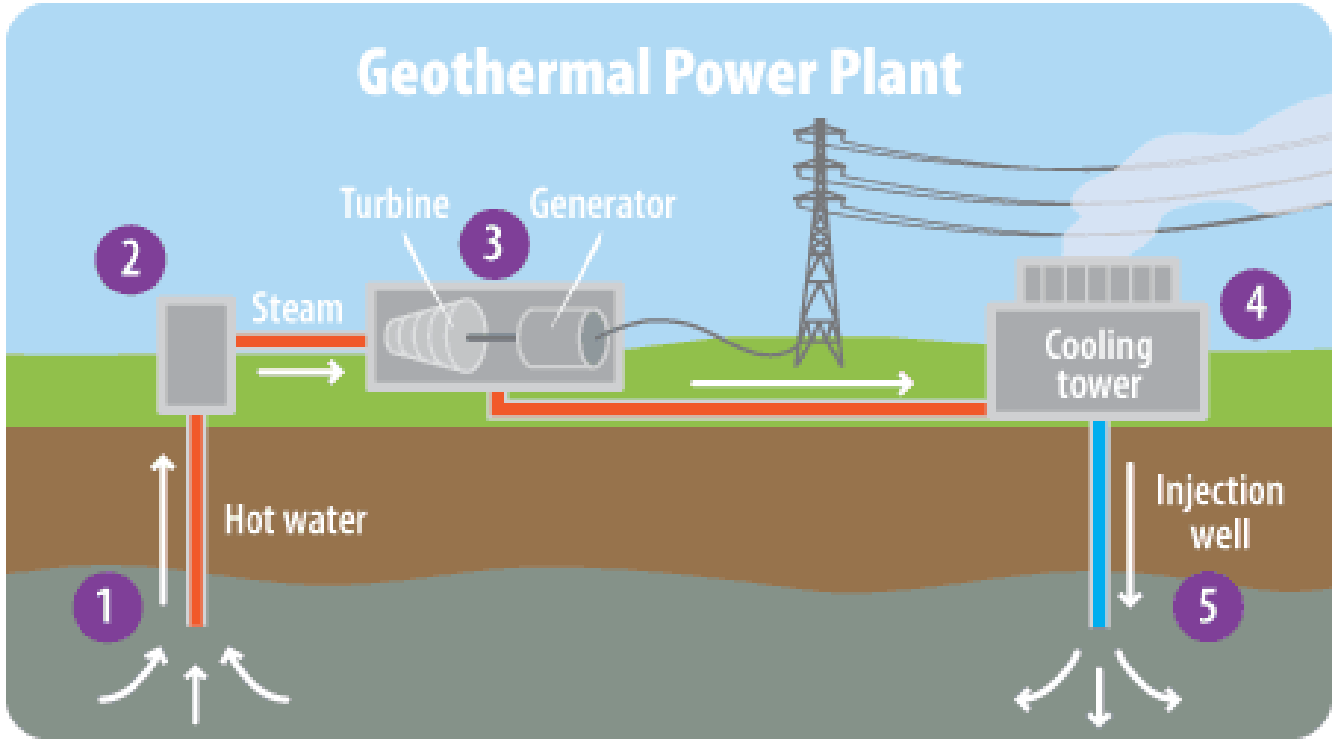
Source: Adapted from the Tennessee Valley Authority (public domain)

Hydropower

- **Advantages**
  - Reliable and consistent power source
  - Low greenhouse gas emissions
  - Multi-purpose use (e.g., irrigation, flood control)
- **Disadvantages**
  - Environmental impact on ecosystems
  - High initial cost of dam construction
  - Limited availability of suitable sites

Geothermal Energy

- Harnesses heat from the **Earth's core** to generate electricity.
  - **Leyte Geothermal Production Field** (1,400 MW)
  - **Makban Geothermal Power Plant**
  - **Tiwi Geothermal Power Plant.**

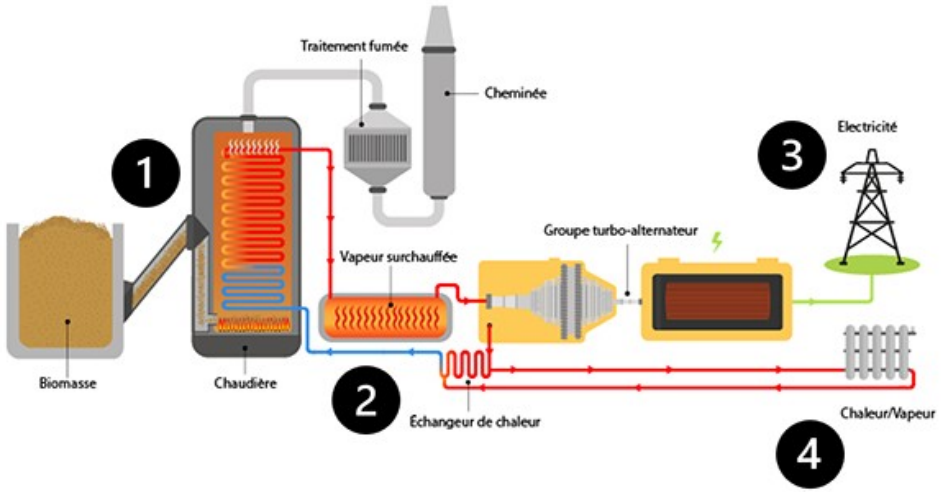


Geothermal Energy

- **Advantages**
  - Low greenhouse gas emissions
  - Reliable and consistent power source
  - Minimal land use
- **Disadvantages**
  - Limited availability of suitable geothermal sites
  - Potential for groundwater contamination
  - High initial cost of installation

Biomass Energy

- **Organic matter**, such as wood, agricultural waste, or animal manure, is burned to produce heat or electricity.



Biomass Energy

- **Advantages**
  - Reduces waste
  - Can be carbon-neutral
  - Versatile fuel source
- **Disadvantages**
  - Potential for air pollution
  - Inefficient energy conversion

Global Energy Challenges

A. Climate Change

- Burning coal, oil, and natural gas releases greenhouse gases that contribute to global warming and climate change.

B. Energy Security

- Countries need a reliable and affordable supply of energy. Dependence on imported fuel can make countries vulnerable to price changes and supply disruptions.

C. Energy Poverty

- Some communities still have limited or unreliable access to electricity and clean energy.

D. Depletion of Fossil Fuels

- Coal, oil, and natural gas are **finite resources**. They take millions of years to form and cannot be replaced quickly.

Cont.

E. Environmental Pollution

- Energy production and consumption can contribute to:
  - Air pollution
  - Water pollution
  - Land degradation
  - Waste generation

F. Increasing Energy Demand

- As population, industries, transportation, and technology grow, the demand for energy also increases.

The Future of Energy

Possible Future Energy Developments

Solar Energy

- Solar technology can be used in homes, schools, and communities.

Electric Transportation

- Electric vehicles can reduce dependence on gasoline and diesel.

Smart Energy Systems

- Technology can help monitor and manage energy consumption more efficiently.

Sustainable Energy Practices

- Turn off lights and appliances when not in use.
- Use energy-efficient appliances.
- Use natural light whenever possible.
- Reduce unnecessary use of air-conditioning.
- Walk, bike, carpool, or use public transportation when possible.
- Use renewable energy when available.
- Maintain electrical equipment properly.
- Reduce, reuse, and recycle materials.
- Plant and protect trees.
- Educate others about responsible energy use.

The future is **not simply about finding more energy**. It is also about using energy more **efficiently** and **responsibly.**

Activity: “Energy: Then, Now, and Tomorrow”

Instructions

- Form 8 groups. Each group will analyze how energy use has changed over time.

Complete the following:

PERIOD	MAIN ENERGY SOURCES	HOW ENERGY WAS/IS USED	PROBLEMS	POSSIBLE SOLUTIONS
Past				
Present				
Future				

Rubric

Criteria	Excellent (4)	Good (3)	Fair (2)	Needs Improvement (1)
Analysis of Energy Sources	Clearly identifies appropriate energy sources for the past, present, and future	Identifies most appropriate energy sources	Identifies some energy sources but with limited explanation	Energy sources are unclear or mostly incorrect
Understanding of Energy Use	Clearly explains how energy is/was/will be used	Explains energy use with minor gaps	Gives limited explanation	Explanation is unclear or incomplete
Problems and Challenges	Clearly identifies relevant problems and challenges	Identifies relevant problems with some explanation	Identifies few problems	Problems are unclear or missing
Possible Solutions	Provides realistic and sustainable solutions	Provides appropriate solutions	Provides limited solutions	Solutions are unrealistic or missing
Group Participation & Presentation	All members participate; presentation is clear and organized	Most members participate; presentation is understandable	Some members participate; presentation needs improvement	

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